PTUK-NLP at the DialectSentEval 2026 Shared Task: MARBERT-Based Sentiment Classification, Missing-Dialect Effects, and Error Analysis

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Abstract

We present our system for Subtask 1 (three-class sentiment classification) of the DialectSentEval 2026 shared task, which evaluates five Arabic dialects using the Multi-Dialect-Sent (MDS-3) hotel-review corpus. Our approach fine-tunes MARBERT with a stratified train/validation split and class-weighted cross-entropy loss. On the official test set, our system achieves an F1 score of 0.8175 and an accuracy of 0.8419, ranking 3rd on the official leaderboard. We identify a substantial dialect mismatch in the dataset: the training set contains no Lebanese reviews, while approximately 20% of the test set is Lebanese. Adding additional Lebanese reviews from OCLAR partially reduces this gap, but does not eliminate it. The error analysis also shows that misclassifications are concentrated in the neutral class and in reviews that express mixed feelings, often with high confidence. These findings highlight the challenges of cross-dialect sentiment classification under unseen-dialect conditions.

1 Introduction

As Arabic-language content on social media and review platforms continues to grow, automatic sentiment analysis (SA) has become increasingly important for applications such as business intelligence (Jakha et al., 2025), public-opinion monitoring (Azmi and Alzanin, 2014), and customer feedback analysis (Alsemaree et al., 2024). However, Arabic SA presents several challenges, including rich morphology, relatively flexible word order, and substantial variation between Modern Standard Arabic (MSA) and regional dialects. While many NLP resources focus on MSA, everyday online communication is dominated by dialects such as Moroccan, Egyptian, Levantine, and Gulf Arabic. Consequently, models trained primarily on MSA or a single dialect may generalize poorly to other dialects, while dialect-specific sentiment resources remain relatively limited (Jaber et al., 2025; Duridi et al., 2025).

The DialectSentEval 2026 shared task addresses this challenge through Subtask 1, which requires three-class sentiment classification of hotel reviews from the Multi-Dialect-Sent (MDS-3) corpus across five Arabic dialects: Moroccan (Darija), Egyptian, Lebanese, Jordanian, and Saudi. This setting provides a practical evaluation of cross-dialect generalization.

We present our MARBERT-based (Abdul-Mageed et al., 2021) system submitted by team PTUK-NLP. The remainder of this paper is organized as follows: 3 reviews related work; 4 describes our system and experimental setup and results; 5 presents our analysis of Lebanese-dialect coverage; 6 discusses the error analysis; and 7 concludes the paper.

2 Task Description

DialectSentEval 2026 (Ezzini et al., 2026a) is a shared task on sentiment analysis and sentiment swapping in Arabic dialects, building on the earlier AraSentEval task (Ezzini et al., 2026b). We participated in on Subtask 1, a three-class sentiment classification task in which systems predict whether a given Arabic sentence expresses positive,

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negative, neutral sentiment. The subtask covers five dialects—Moroccan, Egyptian, Lebanese, Jordanian, and Saudi—and is designed to evaluate robustness to the lexical and syntactic variation across Arabic dialects. Our team, PTUK-NLP, participated in Subtask 1 using the MARBERT-based system described in the following section.

3 Related Work

Arabic sentiment analysis is challenging due to the language's rich morphology and the substantial variation between MSA and regional dialects, which differ in vocabulary, orthography, and syntax. Early approaches relied largely on classical machine-learning methods and handcrafted lexical features, with limited ability to generalize across dialects. The emergence of pretrained Arabic transformers has substantially improved downstream performance. AraBERT (Antoun et al., 2020) was pretrained primarily on MSA and curated Arabic text, while CAMeLBERT (Inoue et al., 2021) introduced separate checkpoints for MSA, dialectal, and mixed-domain pretraining to investigate the effect of corpus composition on downstream performance. MARBERT (Abdul-Mageed et al., 2021), in contrast, was pretrained on a large multi-dialect Arabic Twitter corpus and is therefore particularly suited to informal and dialect-diverse text. Given the dialectal nature of MDS-3, we adopt MARBERT as the backbone of our system.

Several recent studies are closely related to our setting. ADOR (Alharbi et al., 2025c) introduced a human-benchmarked dataset of Arabic dialectal hotel reviews for sentiment analysis, sharing both the hotel-review domain and multi-dialect focus of MDS-3. The AHASIS shared task (Alharbi et al., 2025b) likewise investigated sentiment analysis across Arabic dialects, providing a relevant precedent for our shared-task setting. More recently, Alharbi et al. (2025a) evaluated large language models for Arabic dialect sentiment analysis and showed that dialect coverage in training and prompting data can substantially affect downstream performance. This finding is particularly relevant to our study, where the absence of Lebanese examples from the released training data creates a pronounced train–test dialect mismatch that we investigate in 5.

4 System Description

Figure 1 summarizes our full pipeline, from data preparation through evaluation.

4.1 Data

The released training data (`sent_train_v2.csv`) contains 1,731 labeled sentences, with fields for sentence ID, text, sentiment label, and dialect. The sentiment distribution is negative (657), positive (609), and neutral (465), corresponding to 38%, 35%, and 27% of the data, respectively, indicating a moderate class imbalance. As shown in Figure 2, the training data covers only four of the five target dialects: Darija, Saudi, Jordanian, and Egyptian, with approximately 433 sentences per dialect. Lebanese is absent from the released training data.

We apply minimal preprocessing intended primarily to remove noise while preserving dialect-specific characteristics. Specifically, we remove Arabic diacritics, URLs, and @-mentions, and reduce extreme character elongation by replacing sequences of four or more repeated characters with three characters. We deliberately avoid dialect-to-MSA normalization to retain lexical and orthographic features that may be informative for dialect-specific sentiment classification.

4.2 Model and Training

We fine-tune (UBC-NLP/MARBERT) for three-class sequence classification, using a maximum sequence length of 128 subword tokens. A length audit confirmed that no training examples were truncated, with the 99th percentile at 75 tokens. We use a single stratified 90/10 train/validation split and class-weighted cross-entropy loss with inverse-frequency class weights to address the moderate class imbalance. Training uses AdamW with a learning rate of 2×10^{-5}, batch size 16, weight decay of 0.01, and up to five epochs. Early stopping with a patience of two epochs is applied based on validation macro-F1. No learning-rate warmup or scheduler is used.

4.3 Results and Training Variance

Our submitted configuration achieved an F1 score of 0.8175 and an accuracy of 0.8419 on the official test set, ranking third among participating teams. To assess training variability, we repeated the same configuration twice without code changes. The two additional runs achieved F1 scores of 0.8146

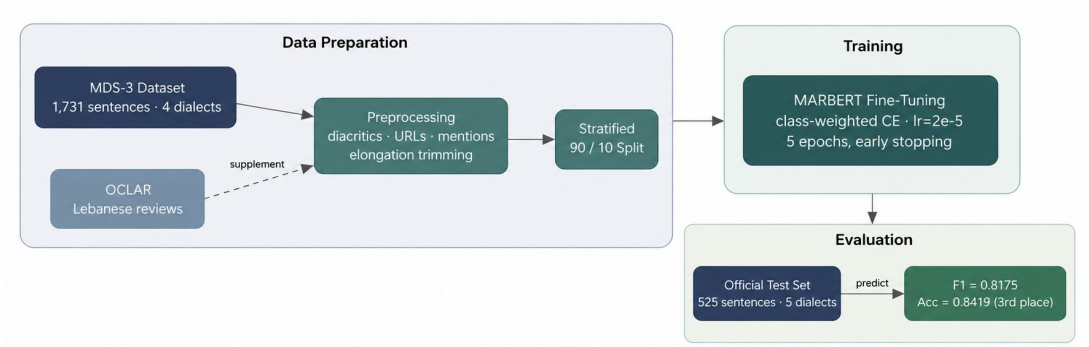


Figure 1: Overview of our system pipeline

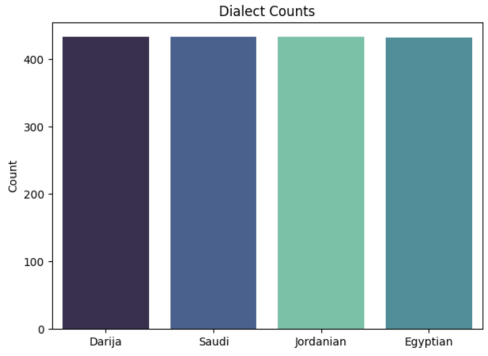


Figure 2: Dialect distribution in the training data. Only four dialects are represented; Lebanese is absent

(accuracy 0.8362) and 0.8031 (accuracy 0.8305). Across the three runs, F1 ranged from 0.8031 to 0.8175, corresponding to a spread of 1.44 F1 points. This variability may arise from stochastic factors during training, including random initialization and GPU-related non-determinism. Given the relatively small training set of 1,731 examples, we report this range to provide a more transparent indication of training variability rather than relying solely on a single run.

5 Investigating the Lebanese Dialect Gap

Although the MDS-3 corpus description specifies five balanced dialects, inspection of the released training data shows that only four are represented: Darija (433), Saudi (433), Jordanian (433), and Egyptian (432). A case-insensitive search of the dialect field found no Lebanese examples. Thus, the training data contains no Lebanese-dialect instances. In contrast, the official test set contains 105 examples from each of the five dialects, meaning that approximately 20% of the test set represents a dialect unseen during training.

Supplementing with OCLAR. To investigate whether additional Lebanese data could alleviate this mismatch, we used the Opinion Corpus for Lebanese Arabic Reviews (OCLAR; Al Omari et al., 2019), a domain-related corpus of Lebanese reviews covering hotel, restaurant, and service experiences. We mapped its 1–5 star ratings to the three sentiment classes as follows: 4–5→positive, 3→neutral, and 1–2→negative. We evaluated two configurations: an unbalanced variant containing 1,771 OCLAR examples, with only the positive class capped, and a dialect-balanced variant capped at 432 examples to approximately match the size of the other MDS-3 dialects. No other system settings were changed.

The unbalanced and balanced configurations achieved official-test F1 scores of 0.791 and 0.792, respectively, both below the original baseline of 0.8175. Notably, the two configurations produced nearly identical results despite the substantially different amounts of supplementary data. This suggests that simply increasing the amount of Lebanese training data is insufficient to overcome the dialect and dataset mismatch.

We further conducted a leave-Lebanese-out diagnostic evaluation in which the model was trained on the four original MDS-3 dialects together with non-held-out OCLAR data and evaluated exclusively on held-out OCLAR examples. The resulting macro-F1 was 0.62, compared with 0.85–0.95 on the four dialects represented in the training data. This substantial difference confirms the difficulty of generalizing to Lebanese Arabic. The limited improvement from OCLAR may partly reflect differences between the two resources, including domain framing (general service reviews versus hotel-specific reviews) and label construction (star ratings versus direct sentiment annotation).

We also evaluated several additional strategies,

including k-fold cross-validation with ensembling, focal loss, back-translation, Arabic orthographic normalization, and automated label-conflict cleaning. None improved over the baseline. Given the limited gains and space constraints, we focus the remainder of the paper on the baseline system and its error analysis.

6 Error Analysis

We analyze the baseline model's errors on the internal validation split (174 sentences). The model achieves a macro-F1 of 0.958, compared with 0.8175 on the official test set, a gap of 14.05 percentage points. This difference is likely influenced by the small validation set and its limited dialect coverage, as it contains no Lebanese examples and therefore does not fully reflect the diversity of the official test set.

6.1 Error Patterns

Table 1 presents the validation confusion matrix. Of the seven errors, six involve the neutral class as either the true or predicted label, indicating that neutral is particularly difficult to distinguish from positive and negative sentiment. This may be partly attributed to its minority representation in the training data (27%) and the semantic ambiguity of reviews containing both positive and negative cues.

	Predicted		
True	pos	neg	neu
positive	60	0	1
negative	1	63	2
neutral	2	1	44

Table 1: Confusion matrix, validation split ($n = 174$).

Qualitative inspection indicates that several errors involve mixed-sentiment reviews, where positive and negative cues occur together. The model tends to rely on a dominant sentiment-bearing expression rather than integrating conflicting cues across the review. For example, a neutral review describing an experience as *"Average, could be better"* while criticizing room and bathroom cleanliness is classified as negative with 99.9% confidence. Similarly, a neutral review praising one receptionist while criticizing other staff is classified as positive with 99.8% confidence.

The model is also highly confident in its incorrect predictions: 171 of 174 predictions (98.3%) have confidence above 0.9, including six of the seven errors. This suggests that the model may be overconfident when processing ambiguous, mixed-sentiment reviews. This observation is consistent with our earlier findings (5), where automated cleaning of label disagreements reduced performance, suggesting that some apparent disagreements may reflect genuine ambiguity rather than annotation noise.

7 Conclusion

Our submitted system is a MARBERT classifier fine-tuned with a single 90/10 stratified split and class-weighted cross-entropy loss, achieving F1 = 0.8175 on the official DialectSentEval 2026 Subtask 1 test set (ranking 3rd on the official evaluation leaderboard), with a measured run-to-run variance of about 1.4 F1 points. We identified and investigated a complete absence of Lebanese-dialect data in the training set; supplementing with a domain-matched Lebanese corpus (OCLAR) narrowed but did not close a substantial generalization gap (0.62 vs. 0.85–0.95 macro-F1). Our error analysis shows the model's failures concentrate on mixed-sentiment reviews labeled neutral, which it misclassifies with high confidence by overweighting a single dominant sentiment cue – a pattern we recommend future work on MDS-3 address directly, for example via explicit modeling of aspect-level or mixed sentiment, rather than through the data-level interventions we found ineffective.

Limitations

Our variance estimate (4.3) is based on three runs of one configuration; a fully rigorous estimate would require more repetitions and should also be established for the OCLAR-augmented variants, whose results in 5 fall within our measured variance range and should not be read as precise rankings. Our diagnosis of the missing-Lebanese-dialect issue is specific to the released `sent_train_v2.csv` file as of our access date. OCLAR differs from MDS-3 in domain scope and label derivation (star ratings vs. direct annotation), limiting how directly its results generalize. Our error analysis is based on a 174-sentence validation split that itself excludes the Lebanese dialect, so it cannot directly characterize Lebanese-specific error patterns.

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